

Process Characterization Report

**A-Mab Drug Substance — Anion Exchange Chromatography (Step 8) —
PCR-008**

Process Development / Manufacturing Science & Technology (MSAT)

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Table of contents

Approvals	3
Abbreviations	4
Executive summary	5
1 Introduction	7
1.1 Product and unit operation	7
1.2 Objectives	7
1.3 Regulatory and scientific basis	8
1.4 Related documents	8
2 Prior knowledge and quality risk basis	10
2.1 Platform and prior-product knowledge	10
2.2 Quality attributes in scope	11
2.3 Risk-based prioritization and parameter selection	11
3 Materials and methods	13
3.1 Scale-down model and its qualification	13
3.2 Operation	13
3.3 Analytical methods	14
3.4 Sampling plan	15
3.5 Statistical methods	16
4 Study design	17
4.1 Factors, ranges and the knowledge space	17
4.2 Screening design	18
4.3 Response-surface design	18
4.4 Univariate assessment	18
5 Results	20
5.1 Centre-point performance and reproducibility	20
5.2 Screening: factor effects	21
5.3 Response-surface models	22
5.4 Mechanistic interpretation	28
6 Design space	30
7 Proven acceptable ranges	31
8 Process capability and robustness	35
9 Parameter classification	37

10 Contribution to the control strategy	39
11 Discussion	40
12 Conclusions	41
13 Deviations from the plan	42
13.1 DEV-008-01 — non-representative load in the first execution	42
13.2 DEV-008-02 — permissive pool collection set-point	43
13.3 DEV-008-03 — equilibration and wash-1 buffer released below its pH range	44
Appendix A — Screening design matrix and responses	45
Appendix B — Response-surface design matrix and responses	47
Appendix C — Full effect and coefficient tables	49
Appendix D — Analytical methods summary	51
References	53

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

AEX, anion exchange chromatography; ANOVA, analysis of variance; BLA, Biologics License Application; CCD, central composite design; CEX, cation exchange chromatography; Cpk, process capability index (one sided); CPP, critical process parameter; CQA, critical quality attribute; CV, coefficient of variation; DNA, deoxyribonucleic acid; DoE, design of experiments; DS, drug substance; ELISA, enzyme-linked immunosorbent assay; GPP, general process parameter; HCP, host cell protein; icIEF, imaged capillary isoelectric focusing; IgG1, immunoglobulin G subclass 1; KPP, key process parameter; LOQ, limit of quantitation; LRF, log reduction factor; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; OLS, ordinary least squares; PAR, proven acceptable range; PRESS, predicted residual error sum of squares; qPCR, quantitative polymerase chain reaction; RSM, response surface methodology; SDM, scale-down model; SOP, standard operating procedure; TCID₅₀, 50 % tissue culture infectious dose; UV, ultraviolet; WC-CPP, well-controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

Executive summary

Anion exchange chromatography is the final polishing step of the A-Mab drug substance process. The step is operated in flow-through mode, and the commercial process runs at 15,000 L. It was characterized against the approved protocol PCP-008 on a qualified scale-down column model, and this report presents the executed study. The report defines the operating region, reports a proven acceptable range for each parameter and each attribute the step is judged on, classifies every parameter, and states what the step contributes to the control strategy.

Of the 5 process parameters of the step, 4 were studied in a multivariate design and 1 was assessed one at a time. The multivariate work consisted of a 19-run two-level screening design and a 28-run face-centred central composite design in the same 4 factors. The responses measured on the flow-through pool were pool host cell protein, the log reduction factors for XMuLV and for MVM, and step yield.

Raising load pH gives the acidic host cell proteins a larger net negative charge and holds more of them on the quaternary amine ligand, and it lowered pool host cell protein by 11.9 ng/mg across the range studied. Raising the conductivity of the equilibration and wash-1 buffer screens that charge and releases the most weakly held of those proteins into the pool, and it raised pool host cell protein by 12.1 ng/mg. Virus particles are held on the same ligand by the acidic groups on their surface, and both log reduction factors fell as load pH fell and as load conductivity rose. The antibody carries a net positive charge at the load pH and is not retained, and no factor changed step yield over the ranges studied. The response-surface models for pool host cell protein and for the two log reduction factors show no significant lack of fit against the reproducibility of the scale-down model (Table 5.6).

Of the characterized region, 85.6% meets every criterion the step is judged against, evaluated on a grid of 14,641 operating points. The host cell protein criterion excludes 14.4% of that region and the two viral criteria exclude no part of it. The normal operating ranges of all 4 multivariate parameters lie inside the characterized region, and for pool host cell protein the range that survives propagation of the other parameters through their normal operating ranges is narrower than the characterization range for two of them (Table 7.1).

At commercial scale the cumulative MVM clearance has a capability of 1.51 against the requirement of 8.6 log₁₀, which is the tightest capability of any A-Mab drug substance attribute. The cumulative XMuLV clearance has a capability of 1.53 and host cell protein a capability of 6.14. This step contributes 4.71 log₁₀ of the MVM claim and 5.95 log₁₀ of the XMuLV claim. The remainder comes from low-pH inactivation (PCR-006) and small-virus retentive filtration (PCR-009), and no part of the cumulative claim rests on this step alone.

All 5 parameters were classified as well-controlled critical process parameters. The study recorded 3 deviations. One of them established that the load material used in the first execution of both designs was not representative of the routine process, and both designs were invalidated and re-executed in full on a requalified load (§13.1). The analysis in this report is the re-executed study throughout. The outcomes summarized here roll up into the master report PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody produced in a fed-batch mammalian cell culture and purified by the Novacyte platform train. The train is Protein A capture, low-pH viral inactivation, cation exchange chromatography, anion exchange chromatography, small-virus retentive filtration and ultrafiltration with diafiltration. Anion exchange is Step 8 and the last chromatographic step of the train. The commercial process is operated at 15,000 L of production bioreactor volume.

The step uses a strong anion exchange resin carrying a quaternary amine ligand and is operated in flow-through mode. At the load pH the antibody carries a net positive charge, because its isoelectric point lies well above the load pH, and it passes through the bed. Species that are negatively charged at that pH bind to the ligand and are removed. These are residual DNA, the acidic fraction of the host cell protein population, leached Protein A, and virus particles, whose surfaces are acidic. The step therefore removes impurities without binding the product, and step yield is high.

Two of the removals are claimed as viral clearance. The step is credited with 5.95 log₁₀ of clearance for the enveloped model virus XMuLV and 4.71 log₁₀ for the non-enveloped model virus MVM, and both claims are modular in the sense of ICH Q5A(R2) (International Council for Harmonisation 2023a). The MVM capsid has no lipid envelope, so the low-pH hold at Step 6 leaves the particle infectious and the cumulative MVM requirement is met by the binding of MVM to the ligand here and its retention by the virus filter at Step 9. The MVM attribute is registered against this step in the drug substance quality attribute register (Table 2.1). The step also clears host cell protein, residual DNA and leached Protein A, all of which are formed or introduced upstream and are registered against the steps that form them (Table 2.2).

No aggregate clearance and no charge-variant modification are claimed for this step. Aggregate is controlled at cation exchange (PCR-007) and the charge-variant distribution is set in the production bioreactor (PCR-003).

1.2 Objectives

The objectives of the study reported here were the following.

- (i) Quantify the relationship between each studied process parameter and each quality attribute and performance attribute measured at the step, over ranges wider than the intended normal operating ranges.

- (ii) Define the operating region within which the step meets every criterion it is judged against, and identify the attribute that bounds it.
- (iii) Establish a proven acceptable range for each parameter against each attribute that carries an acceptance criterion at this step, both alone and with the remaining parameters varying within their normal operating ranges.
- (iv) Classify each parameter according to the decision logic of SOP-4001 and the criticality framework of the A-Mab control strategy (CMC Biotech Working Group 2009).
- (v) Justify the ranges to be run in Stage 2 and state what the step contributes to the control strategy, including the in-process controls that follow from the study.

1.3 Regulatory and scientific basis

The study is a Stage 1 process design activity in the sense of the FDA process validation guidance (U.S. Food and Drug Administration 2011). It follows the enhanced approach of ICH Q8(R2) and ICH Q11 (International Council for Harmonisation 2009, 2012), in which prior knowledge and a risk assessment direct the experimental work, and the experimental work then puts limits on a process that prior knowledge already describes. Criticality is treated as a continuum in the sense of ICH Q9 (International Council for Harmonisation 2023b), so an attribute is assigned a level of criticality and the levels called high and very high are called critical. The viral clearance claims follow ICH Q5A(R2) (International Council for Harmonisation 2023a), which allows the clearance of a process to be built from independently studied modules. The process model, the quality attribute framework and the risk methodology follow the A-Mab case study (CMC Biotech Working Group 2009).

Scale-down models were qualified under SOP-1001 and the designed experiments were designed, executed and analysed under SOP-1002. The step itself was operated under SOP-2011 and the resin was packed, sanitized and life-cycled under SOP-2008. Parameter classification follows SOP-4001.

1.4 Related documents

This report is one of the per-step reports of the A-Mab characterization campaign. It executes the protocol PCP-008, takes its scope from the pre-characterization risk assessment RA-001, and rolls up into the master report PCMR-001.

Table 1.1: Related documents.

Docu- ment ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope

Docu- ment ID	Title	Relationship
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-008	Process Characterization Plan (Anion Exchange Chromatography (Step 8))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

Anion exchange chromatography in flow-through mode has been used for three previously licensed Novacyte antibodies (X-Mab, Y-Mab and Z-Mab) and throughout A-Mab clinical manufacture. The resin, the ligand chemistry, the bed height and the buffer system are the platform ones, and the operating conditions have not changed materially between these products. The body of experience shows that the step removes acidic impurities and virus particles reproducibly and that the product recovery is high.

The load pH fixes the charge each species in the feed carries relative to its isoelectric point. Raising it makes the acidic host cell proteins, the DNA and the virus particles more negative and holds them more tightly on the quaternary amine ligand, while the antibody stays positive and flows through, so a higher load pH was expected to lower pool host cell protein and to raise both log reduction factors. The equilibration and wash-1 buffer fills the bed while the bound species are held there, and the ions it carries screen the attraction between those species and the ligand. Raising its conductivity was therefore expected to release the species that carry the least charge, which is the opposite direction to a rise in load pH. The load carries ions of its own, and raising the load conductivity screens the same attraction while the impurities and the virus particles are still competing for the ligand, so it was expected to lower removal as well.

Protein load in flow-through mode is the mass of antibody passed per volume of resin. What reaches the pool with it is the impurity mass it carries, because the ligand holds a finite mass of bound species and a load heavy enough to approach that mass lets host cell protein break through. Within the range studied here the impurity mass carried onto the column is far below the ligand capacity, so protein load was expected to have little effect. Raising the operating flow rate shortens the time the fluid spends in the bed and leaves a species less time to reach the ligand. Host cell protein, DNA and virus particles are small and highly charged and bind within a fraction of the residence time this range spans, so a higher flow rate was expected to lower removal only slightly.

Two further expectations come from prior product experience. Residual DNA is a polyanion at any pH the step could be operated at, so it binds the anion exchanger strongly and is cleared by orders of magnitude irrespective of the operating point. Leached Protein A is acidic and binds the ligand while the antibody passes through, so the step removes the ligand fragments introduced by the capture step. Neither attribute was carried into the designed experiments, for the reasons given in §3.3.

2.2 Quality attributes in scope

One drug substance attribute is registered against this step. It is the cumulative clearance of MVM, the non-enveloped model virus, and it is a very high criticality attribute of the viral safety category.

Table 2.1: Drug substance quality attribute set by this step.

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — MVM (parvovirus)	viral safety	8.6-20 log10 (cumulative)	VH	20

The criticality levels in Table 2.1 and Table 2.2 come from the A-Mab attribute assessment, in which the impact of an attribute on safety and efficacy is combined with the uncertainty in that impact to give the Tool #1 score (CMC Biotech Working Group 2009). Attributes scored high or very high are treated as critical. The assessment scores MVM clearance at the highest severity in the register, because no later step and no release test compensates for virus that leaves this step in the pool.

Four further attributes are cleared but not formed at this step, and they are listed in Table 2.2 with the same columns. Their acceptance criteria apply to the drug substance, so this step is judged against the fraction of each criterion that must be met at its own outlet, as described in §3.5.

Table 2.2: Drug substance quality attributes cleared at this step.

CQA	Category	Acceptance	Criticality	Tool #1
Host Cell Protein (HCP)	process impurity	0-100 ng/mg	M-H	36
Residual DNA	process impurity	0-0.001 ng/dose	VL	6
Leached Protein A	process impurity	0-5 ppm	L-M	16
Viral clearance — XMuLV (enveloped)	viral safety	16.7-30 log10 (cumulative)	VH	20

Host cell protein and the two viral attributes were measured in the designed experiments. Residual DNA and leached Protein A were not, and §3.3 gives the reason.

2.3 Risk-based prioritization and parameter selection

The scope of this study was set by the pre-characterization risk assessment RA-001, which ranked every parameter of the step by its potential impact on a quality attribute and by its potential to interact with other parameters, and then filtered the ranked

list into study types. Parameters that scored high on impact, on interaction potential or on both were assigned to the multivariate designs. Parameters that scored high on impact but low on interaction potential were assigned to univariate assessment. Parameters whose ranges are set by the equipment or by the platform and which carry no plausible route to a quality attribute were not studied, and their ranges are held at the platform values.

The assessment carried 4 parameters into the multivariate designs. These are load pH, the equilibration and wash-1 conductivity, load conductivity and protein load. Load pH changes the charge the acidic impurities and the virus particles carry, the two conductivities change the ionic strength that screens that charge, and protein load changes the mass of impurity competing for the ligand. Each of the four therefore changes how tightly those species are held, so the effect of one was expected to depend on the level of another, and a multivariate design was needed to estimate that dependence. Operating flow rate was assigned to univariate assessment, because its route to an attribute is the residence time and because that route does not depend on the level of the other parameters.

Buffer identity, buffer molarity, bed height, column diameter, sanitization regime and resin lot are controlled to the platform specification and were not characterized. They are subject to identity and quality control on receipt and to the resin life-cycle controls of SOP-2008. Their exclusion is a decision recorded in RA-001 and is not revisited here.

3 Materials and methods

3.1 Scale-down model and its qualification

The study was executed on a scale-down column model of the commercial step. The model uses the same resin, the same ligand chemistry and the same buffer system as the commercial column, and it is scaled on the established principles of chromatographic scaling. Bed height, linear flow velocity, protein load per volume of resin, and the equilibration, wash and pool volumes normalized to column volumes are held equal to the commercial values, so that the residence time and the number of column volumes each species experiences are the same at the two scales. Column packing was verified by plate count and peak asymmetry before each campaign under SOP-2008. The column was operated in the temperature-controlled chromatography chamber EQ-CHR-118, which was within its calibration interval throughout execution.

The model was qualified under SOP-1001 by operating it at the commercial set-points and comparing its input and output attributes with the corresponding at-scale data. The attributes compared were product recovery, pool protein concentration, pool host cell protein and the pool volume expressed in column volumes. No statistically significant difference between scales was found, and the qualification record is maintained under SOP-1001. On that basis the conclusions drawn from the model in this report are applied to the commercial step.

Reproducibility of the model within a design is measured by its centre-point replicates. Screening carried 3 centre points and the response-surface design carried 4, and these replicates provide the pure-error term against which lack of fit is tested in §5.3. Their dispersion is reported in §5.1.

Two limits apply to the model. It reproduces the chemistry of the step and not the packing variability of a commercial column, so column performance at scale is verified by plate count and asymmetry at each packing under SOP-2008, and not by this study. It also uses resin within the early part of its life cycle, so the effect of cumulative cycles on clearance is covered by the resin life-cycle programme of SOP-2008 and not by the models fitted here.

3.2 Operation

The step is operated in flow-through mode under SOP-2011. The column is equilibrated with the equilibration buffer, whose conductivity is one of the studied parameters. The load is the cation exchange eluate from Step 7, adjusted to the target load pH and to the target load conductivity before it is applied. The product passes through the

bed while the acidic species bind, and the flow-through is collected. A wash with the equilibration buffer follows the load and recovers the product held in the interstitial volume, and the wash is collected into the same pool.

Pool collection starts and stops on the ultraviolet absorbance of the column effluent. The trailing-edge stop set-point used in both executions of the study was 180 mAU. Reconciliation of the pool fractions after execution showed that this set-point was slightly permissive, and it was corrected to 120 mAU. The investigation, the impact assessment and the verification runs are in §13.2.

The load material for the executed study was the requalified cation exchange eluate LOT-CEX-8851, whose charge-variant profile was verified against the routine process before the designs were run. The load material used in the first, invalidated execution was LOT-CEX-8842, and §13.1 gives the reason it was not representative.

Buffer identity and molarity, resin lot, bed height and column diameter are controlled to the platform specification. They are subject to identity and quality control on receipt and to the life-cycle controls of SOP-2008, and they were held at their platform values in every run. They are therefore controlled inputs to this study and not characterized parameters.

3.3 Analytical methods

Each response was measured by a validated method. Pool host cell protein was measured by the host cell protein enzyme-linked immunosorbent assay AMV-3012. XMuLV and MVM titres were measured by infectivity assay under AMV-3017 and AMV-3018, and the log reduction factor of a run is the difference between the titre applied to the column and the titre recovered in the pool, each expressed per unit volume and corrected for the volumes involved. Step yield was calculated from the protein concentration of the load and of the pool, measured by ultraviolet absorbance at 280 nm, and from the recorded volumes. The charge-variant profile of the load material was measured by imaged capillary isoelectric focusing under AMV-3013, and residual DNA and leached Protein A are measured under AMV-3014 and AMV-3016. The controlled documents for the step and its methods are listed in Table 3.1.

Table 3.1: Controlled documents for this step.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2011	Operation of the Anion-Exchange Polishing Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP

Reference	Title	Type
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

Method performance bounds what the study can resolve. The host cell protein assay has an intermediate precision of 9.5 % relative standard deviation and a limit of quantitation of 1 ng/mg, and about 40.0% of the variance observed on a host cell protein result is attributable to the assay. The infectivity assays are more precise on the log10 scale, at 0.3 % for XMuLV and 0.32 % for MVM, with limits of quantitation of 1.5 and 1.4 log10 TCID50 per mL. Full method performance is given in Appendix D.

Residual DNA and leached Protein A were measured on the pools but were not carried into the designed experiments as responses. Two reasons apply. DNA is a polyanion at every pH the step can be operated at and binds the anion exchanger under every condition in the design, so no parameter dependence was expected within the ranges studied. Both attributes also sit far below their drug substance criteria at the outlet of the process, at 0.0001 ng per dose against a criterion of 0.001 ng per dose and 0.003 ppm against a criterion of 5 ppm (§8). A designed experiment on either attribute would have resolved differences that carry no consequence for the drug substance.

3.4 Sampling plan

One load sample and one pool sample were taken from every run of both designs. The load sample was assayed for protein concentration, host cell protein and charge variants, and the pool sample for protein concentration, host cell protein, residual DNA and leached Protein A. Every design run was executed as a virus spiking run on the qualified small-scale spiking model, so XMuLV and MVM titres were measured on the load and on the pool of every run. Spiking studies are performed in a containment laboratory on a separate column of the same scale-down design, as required for a modular clearance claim under ICH Q5A(R2) (International Council for Harmonisation 2023a).

Centre points were distributed through the run order of each design and not grouped together, so that a drift in the model or in an assay over a campaign would appear as dispersion between replicates. The screening design carried 3 centre points and

the response-surface design carried 4. Run order within each design was randomized under SOP-1002.

3.5 Statistical methods

Every factor was coded so that the edges of its characterization range are -1 and $+1$ and the midpoint of that range is 0 . Coding puts the factors on a common scale, so a coefficient is the change in the response over half the characterization range of its factor and coefficients can be compared across factors of different units. An effect in a screening table is twice the coded coefficient, which is the change in the response between the low and the high edge of the range.

The screening designs were analysed by ordinary least squares with all main effects and all two-factor interactions, which is 11 terms including the intercept. The response-surface designs were analysed with the full quadratic model, which adds the squared term of each factor and gives 15 terms. A term was treated as significant at $\alpha = 0.05$. Model adequacy was assessed from the coefficient of determination, its adjusted form, the predicted form computed from PRESS by leave-one-out deletion, the model F ratio and its p value, and the analysis of variance partition of the residual into lack of fit and pure error. The pure-error term is the dispersion between the centre-point replicates, so a significant lack of fit means the model misses structure that is larger than the reproducibility of the model itself.

The screening design identifies which factors matter and gives a first estimate of their size. The response-surface model is the predictive model of this report. Every prediction, every proven acceptable range and every statement about the operating region is taken from the response-surface models and not from the screening fits.

Three criteria are used to judge the step. Pool host cell protein is judged against an in-process limit of 21.7 ng/mg. That limit was obtained by carrying the drug substance criterion of 100 ng/mg back through the host cell protein clearance credited to the steps that follow this one, which gives the pool concentration a batch may carry and still reach the criterion, and by then setting the limit a factor below that concentration. A limit set at the concentration itself would hold only while every later step delivered its nominal clearance exactly. The two viral responses are judged against the step contribution the cumulative claim requires. The cumulative MVM requirement is 8.6 log₁₀ and the other steps of the train are credited with 5.32 log₁₀, so this step must deliver 3.28 log₁₀. The same calculation for XMULV gives a required contribution of 3.78 log₁₀. Step yield has no acceptance criterion at this step and no proven acceptable range is computed for it.

Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches of the whole train. Each batch draws every parameter of every step from a normal distribution centred on its set-point with a standard deviation of one sixth of the width of its normal operating range, truncated at the edges of the characterization range. Capability is reported as a one-sided Cpk, computed against the upper limit for an impurity and against the lower limit for a clearance requirement, because a one-sided attribute is not penalized by its distance from a limit that does not apply to it.

4 Study design

4.1 Factors, ranges and the knowledge space

Table 4.1 gives the five parameters of the step with their set-points, their normal operating ranges, the ranges over which they were characterized, the study type each received and the classification each was given by this study. The classification column is the outcome of §9 and is repeated here so that the parameter set can be read in one place.

Table 4.1: Process parameters, ranges, study type and classification.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Load pH	pH	7.5	7.35–7.65	7.2–7.8	WC-CPP	multivariate
Equil/Wash-1 conductivity	mS/cm	2.6	2.1–3.1	1.6–3.6	WC-CPP	multivariate
Load conductivity	mS/cm	5.5	4–7	3–8	WC-CPP	multivariate
Protein load	g/L resin	200	120–280	50–300	WC-CPP	multivariate
Operating flow rate	cm/hr	300	200–400	150–450	WC-CPP	univariate

The characterization ranges are deliberately wider than the normal operating ranges. Load pH was studied from 7.2 to 7.8 against a normal operating range of 7.35 to 7.65, and the equilibration and wash-1 conductivity from 1.6 to 3.6 mS/cm against a normal operating range of 2.1 to 3.1 mS/cm. Protein load was studied from 50 to 300 g per L of resin, which reaches well above the upper edge of its normal operating range of 280 g per L. Studying wider ranges serves two purposes. It exposes the step to conditions it would meet only under a combined excursion, and it leaves room for movement of the operating point within a characterized region without new experimental work.

The characterized region is the knowledge space of the step. It is the four-dimensional box whose edges are the characterization ranges in Table 4.1. The design space is the part of that box in which every criterion is met, and it is reported in §6. The normal operating ranges lie inside both.

The four multivariate factors are coded A to D in the effect and coefficient tables of §5 and of Appendix C. Table 4.2 gives the mapping.

Table 4.2: Coded factor legend.

Code	Factor	Unit	Studied in
A	Load pH	pH	screening + RSM
B	Equil/Wash-1 conductivity	mS/cm	screening + RSM
C	Load conductivity	mS/cm	screening + RSM
D	Protein load	g/L resin	screening + RSM

4.2 Screening design

The screening design was a two-level full factorial in the 4 multivariate factors, 16 factorial runs augmented with 3 centre points, for 19 runs in total. The full factorial was run in preference to a fractional design because each of the four factors changes how tightly the same acidic species are held on the same ligand, so an interaction between any two of them was expected. In a full factorial no main effect is aliased with a two-factor interaction and every two-factor interaction is estimated in its own right, so an interaction found in this design is a property of the data and not of the aliasing structure. The design supports 11 model terms and leaves 8 residual degrees of freedom. The coded matrix and the measured responses are in Appendix A.

4.3 Response-surface design

The response-surface design was a face-centred central composite design in the same 4 factors. It comprises 16 factorial runs, 8 axial runs and 4 centre points, for 28 runs. The axial points sit on the faces of the coded cube and not outside it, so every run of the design is inside the characterization ranges of Table 4.1 and no prediction from the model requires extrapolation beyond a range the step was operated over. The design adds the four squared terms to the screening model and so allows curvature in each factor to be estimated. It leaves 13 residual degrees of freedom, of which 3 are pure error. The coded matrix and the measured responses are in Appendix B.

The same four factors were carried from screening into the response surface. Screening found large effects on host cell protein for load pH and the equilibration and wash-1 conductivity, and large effects on the two log reduction factors for load pH and load conductivity, so dropping any factor would have lost a term that carries one of the three attributes.

4.4 Univariate assessment

Operating flow rate was assessed one factor at a time over the range in Table 4.1. It changes only the time a species has to reach the ligand before the fluid moves on, and it changes neither the charge that species carries nor the ionic strength that screens

it, which is why RA-001 ranked its potential to interact with the other parameters low and assigned it to a univariate assessment. No effect estimate for flow rate is reported in §5, because it was not a factor of either design, and its classification in §9 rests on the univariate assessment together with the control capability of the pump.

5 Results

5.1 Centre-point performance and reproducibility

The centre-point replicates measure the reproducibility of the scale-down model together with the assays, at one operating point, within one campaign. Table 5.1 gives the replicates of the screening design and Table 5.2 those of the response-surface design.

Table 5.1: Centre-point reproducibility, screening design.

Response	n	Mean	SD	%CV
Pool HCP (ng/mg)	3	15.1	3.11	20.6
XMuLV LRF ($\log_{10}$)	3	6.39	0.426	6.67
MVM LRF ($\log_{10}$)	3	4.81	0.268	5.57
Step yield	3	0.972	0.00523	0.538

Table 5.2: Centre-point reproducibility, response-surface design.

Response	n	Mean	SD	%CV
Pool HCP (ng/mg)	4	13.7	2.48	18.2
XMuLV LRF ($\log_{10}$)	4	6.63	0.445	6.71
MVM LRF ($\log_{10}$)	4	4.9	0.336	6.86
Step yield	4	0.966	0.0123	1.27

Pool host cell protein is the least precise response in both designs, at 20.6 % in screening and 18.2 % in the response-surface design. Much of that dispersion is analytical. The host cell protein assay carries an intermediate precision of 9.5 % and accounts for about 40.0% of the variance of a result, so the process contribution to the observed spread is smaller than the total. The two viral responses reproduce at 6.7 % and 6.9 % of their means, which on the $\log_{10}$ scale is a standard deviation of 0.45 and 0.34 $\log_{10}$. Step yield reproduces at 1.3 %.

The centre points of the two designs agree with each other. Pool host cell protein averaged 15.1 ng/mg in screening and 13.7 ng/mg in the response-surface design, a difference well inside the dispersion of either set. Both designs were executed on the same requalified load material, and their agreement supports reading the screening and the response-surface results as one study.

5.2 Screening: factor effects

Table 5.3 summarizes the screening models. Three of the four responses gave a significant model and step yield did not.

Table 5.3: Screening model summary, all responses.

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
Pool HCP (ng/mg)	19	0.966	0.923	0.841	22.6	8.55e-05	2.45
XMULV LRF (log ₁₀)	19	0.842	0.644	-0.106	4.25	0.0257	0.393
MVM LRF (log ₁₀)	19	0.97	0.932	0.869	25.8	5.18e-05	0.178
Step yield	19	0.498	-0.13	-3.64	0.792	0.642	0.0126

Pool host cell protein was affected by the equilibration and wash-1 conductivity, by load pH and by the interaction between the two. Table 5.4 gives the effect estimates in order of size. Raising the equilibration and wash-1 conductivity from 1.6 to 3.6 mS/cm raised pool host cell protein by 12.1 ng/mg ($p = 9\text{e-}06$). Raising load pH from 7.2 to 7.8 lowered it by 11.9 ng/mg ($p = 1\text{e-}05$). The two effects are of similar size and opposite sign. Moving load pH across its range binds about as much host cell protein as moving the buffer conductivity across its own range releases. The interaction between them was -6.6 ng/mg ($p = 0.0006$) and is negative, so raising load pH removes more host cell protein when the conductivity is high than when it is low. Neither load conductivity nor protein load reached significance for this response.

Table 5.4: Screening effect estimates, pool host cell protein.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	12.1	6.04	0.613	9.85	9.48e-06	***
A	-11.9	-5.96	0.613	-9.72	1.05e-05	***
AB	-6.63	-3.32	0.613	-5.41	0.000642	***
C	-1.66	-0.829	0.613	-1.35	0.213	
AC	1.65	0.827	0.613	1.35	0.214	
BC	-1.21	-0.605	0.613	-0.986	0.353	
D	0.885	0.443	0.613	0.722	0.491	
AD	-0.498	-0.249	0.613	-0.406	0.696	
CD	0.261	0.13	0.613	0.212	0.837	
BD	0.224	0.112	0.613	0.182	0.86	

The two viral responses were affected by load pH and by load conductivity, and by nothing else. For MVM, shown in Table 5.5, load pH raised the log reduction factor by 1.22 log₁₀ across its range ($p = 8\text{e-}07$) and load conductivity lowered it by 0.71 log₁₀ ($p = 4\text{e-}05$). The XMULV response behaved the same way, with an effect of 1.00

log10 for load pH ($p = 0.0009$) and 0.60 log10 for load conductivity ($p = 0.02$). The equilibration and wash-1 conductivity, which dominates the host cell protein response, did not reach significance for either virus. No interaction reached significance for either viral response in this design.

Table 5.5: Screening effect estimates, MVM log reduction factor.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	1.22	0.61	0.0446	13.7	7.83e-07	***
C	-0.715	-0.357	0.0446	-8.01	4.32e-05	***
B	0.182	0.091	0.0446	2.04	0.0757	
BD	0.107	0.0536	0.0446	1.2	0.263	
AD	-0.0627	-0.0314	0.0446	-0.704	0.502	
D	-0.0451	-0.0226	0.0446	-0.506	0.626	
BC	-0.0442	-0.0221	0.0446	-0.496	0.634	
CD	0.0334	0.0167	0.0446	0.375	0.718	
AC	0.0135	0.00677	0.0446	0.152	0.883	
AB	-0.0132	-0.00659	0.0446	-0.148	0.886	

Step yield gave no significant model, at $F = 0.79$ on $p = 0.64$, and no individual term reached $\alpha = 0.05$. The largest estimate was that of load conductivity, at -0.012 of the fraction recovered, which is smaller than the dispersion between the centre points of the same design. The step recovers the product irrespective of where it is operated within these ranges.

The predicted coefficient of determination in Table 5.3 is negative for XMuLV, at -0.11. A screening design with 8 residual degrees of freedom and ten estimated terms leaves little room for a deletion test, and this value says that the screening fit for XMuLV should not be used to predict an unrun condition. It is used here only to identify which factors act, and the response-surface model of §5.3 carries the prediction.

The screening designs identify the factors and give a first estimate of their size. The estimates are refined by the response-surface models, and where the two disagree in magnitude the response-surface estimate is the one used.

5.3 Response-surface models

Table 5.6 summarizes the response-surface models. Pool host cell protein and the two log reduction factors gave significant models with the largest part of their variation explained, and step yield again gave no model.

Table 5.6: Response-surface model summary, all responses.

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
Pool HCP (ng/mg)	28	0.934	0.864	0.66	13.2	1.83e-05	2.65
XMuLV LRF (log ₁₀)	28	0.915	0.824	0.621	10	8.69e-05	0.314
MVM LRF (log ₁₀)	28	0.898	0.787	0.394	8.14	0.000269	0.32
Step yield	28	0.488	-0.0632	-2.18	0.885	0.589	0.0145

The host cell protein model explains 93.4% of the variation, with an adjusted value of 86.4% and a predicted value of 66.0%. Its residual standard deviation is 2.65 ng/mg, which is of the same order as the centre-point standard deviation in Table 5.2. Table 5.7 gives the coefficients. Load pH and the equilibration and wash-1 conductivity carry the response, at -5.34 and 6.19 ng/mg per half range, and their interaction is retained at -1.45 (p = 0.05). No squared term reached significance, so the surface is close to a plane over the characterization range. Load conductivity and protein load have coefficients of 0.04 and 0.39 ng/mg per half range, neither of which is distinguishable from zero.

Table 5.7: Response-surface coefficients, pool host cell protein.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	14.8	0.917	16.2	5.51e-10	***
A	-5.34	0.625	-8.54	1.09e-06	***
B	6.19	0.625	9.89	2.04e-07	***
C	0.0399	0.625	0.0637	0.95	
D	0.387	0.625	0.619	0.546	
AB	-1.45	0.663	-2.19	0.0475	*
AC	-0.925	0.663	-1.39	0.187	
AD	0.0192	0.663	0.029	0.977	
BC	0.15	0.663	0.226	0.825	
BD	-0.559	0.663	-0.843	0.415	
CD	-0.124	0.663	-0.187	0.855	
A ²	1.91	1.65	1.16	0.268	
B ²	2.22	1.65	1.34	0.202	
C ²	-0.0187	1.65	-0.0113	0.991	
D ²	-1.98	1.65	-1.2	0.253	

The MVM model explains 89.8% of the variation and the XMuLV model 91.5%. Their coefficients are in Table 5.8 and Table 5.9. Load pH raises the MVM log reduction factor by 0.54 log₁₀ per half range and load conductivity lowers it by 0.56 log₁₀ per half range, and the two coefficients are almost equal in size. The XMuLV model has the same pair of terms with almost the same coefficients. One interaction, between

load pH and protein load, reached significance for XMuLV at 0.19 log10 ($p = 0.03$). It is about a third of the load pH main effect and it is the only significant interaction among the two viral responses in either design. With 15 terms in the model, a term of that p value is expected about once in twenty by chance. The region defined in §6 is bounded by pool host cell protein and by no viral response, so it is the same region whether this term is kept or dropped.

Table 5.8: Response-surface coefficients, MVM log reduction factor.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	4.85	0.111	43.9	1.63e-15	***
A	0.539	0.0754	7.15	7.48e-06	***
B	0.0307	0.0754	0.407	0.691	
C	-0.564	0.0754	-7.48	4.65e-06	***
D	0.0261	0.0754	0.346	0.735	
AB	0.127	0.08	1.59	0.136	
AC	-0.074	0.08	-0.925	0.372	
AD	0.0353	0.08	0.441	0.666	
BC	-0.0339	0.08	-0.424	0.679	
BD	0.00805	0.08	0.101	0.921	
CD	-0.0156	0.08	-0.195	0.848	
A ²	-0.322	0.199	-1.61	0.13	
B ²	0.149	0.199	0.747	0.469	
C ²	0.0224	0.199	0.112	0.912	
D ²	0.135	0.199	0.679	0.509	

Table 5.9: Response-surface coefficients, XMuLV log reduction factor.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	6.57	0.109	60.6	2.51e-17	***
A	0.621	0.074	8.39	1.33e-06	***
B	-0.076	0.074	-1.03	0.323	
C	-0.554	0.074	-7.48	4.6e-06	***
D	0.0287	0.074	0.388	0.704	
AB	-0.0278	0.0785	-0.355	0.729	
AC	0.0223	0.0785	0.285	0.78	
AD	0.192	0.0785	2.44	0.0295	*
BC	-0.118	0.0785	-1.5	0.158	
BD	0.0577	0.0785	0.735	0.475	
CD	-0.0751	0.0785	-0.957	0.356	
A ²	0.0696	0.195	0.356	0.727	
B ²	-0.328	0.195	-1.68	0.118	
C ²	0.152	0.195	0.779	0.45	
D ²	0.0761	0.195	0.39	0.703	

The predicted coefficient of determination is 39.4% for MVM and 62.1% for XMuLV. The

MVM value is the weakest of the three responses that carry an acceptance criterion, and it says that a run deleted from the design is predicted with appreciably more error than the fit within the design suggests. The margin between the predicted clearance and the required contribution is large enough to absorb that error at every point of the characterized region (§6), and the statement made from this model is a statement about the mean level of clearance and not about an individual run.

Lack of fit was tested against the pure error of the centre-point replicates for each of the three responses that carry an acceptance criterion. Table 5.10, Table 5.11 and Table 5.12 give the partitions. Lack of fit is not significant for any of the three, at $p = 0.50$ for host cell protein, $p = 0.91$ for XMuLV and $p = 0.62$ for MVM. The quadratic model therefore misses no structure that is large compared with the reproducibility of the scale-down model itself.

Table 5.10: Analysis of variance with lack of fit, pool host cell protein.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	1,302.7	14	93.1	13.2	1.83e-05
Residual	91.5	13	7.04	nan	nan
Lack of fit	73.0	10	7.3	1.19	0.499
Pure error	18.5	3	6.16	nan	nan

Table 5.11: Analysis of variance with lack of fit, XMuLV log reduction factor.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	13.8	14	0.988	10	8.69e-05
Residual	1.28	13	0.0985	nan	nan
Lack of fit	0.687	10	0.0687	0.347	0.911
Pure error	0.594	3	0.198	nan	nan

Table 5.12: Analysis of variance with lack of fit, MVM log reduction factor.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	11.7	14	0.834	8.14	0.000269
Residual	1.33	13	0.102	nan	nan
Lack of fit	0.994	10	0.0994	0.881	0.619
Pure error	0.339	3	0.113	nan	nan

Figure 5.1 shows the fitted surfaces over load pH and the equilibration and wash-1 conductivity, and Figure 5.2 shows them over load pH and load conductivity. The first pair of axes carries the host cell protein response, whose contours run diagonally across the panel and reach their highest values at low load pH with high conductivity. The second pair carries the two viral responses, whose contours run in the opposite sense, with clearance highest at high load pH and low load conductivity. In both panels the step yield surface is flat.

Response-surface predictions (remaining factors held at set-point)

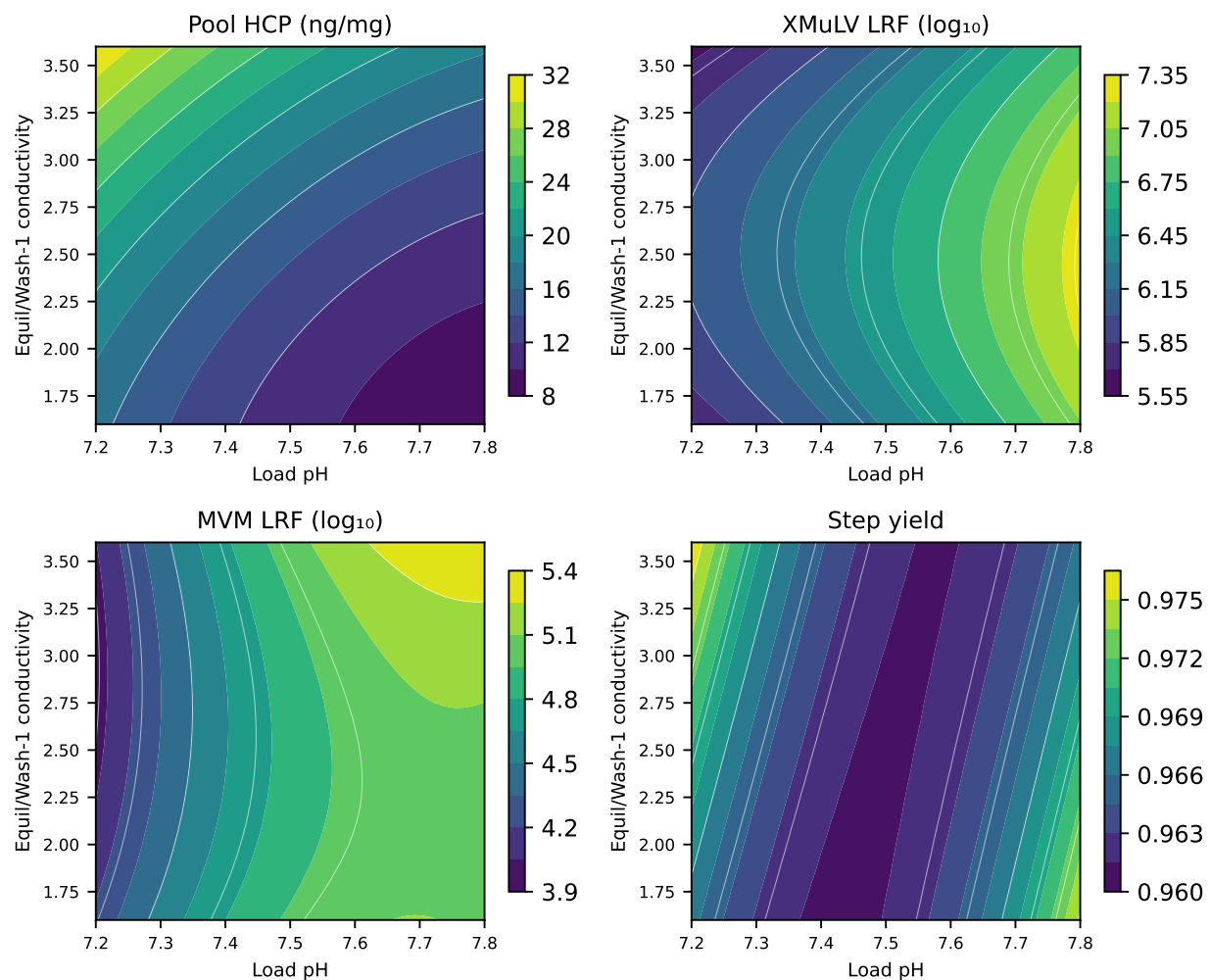


Figure 5.1: Fitted response surfaces over load pH and the equilibration and wash-1 conductivity, one panel per response.

Response-surface predictions (remaining factors held at set-point)

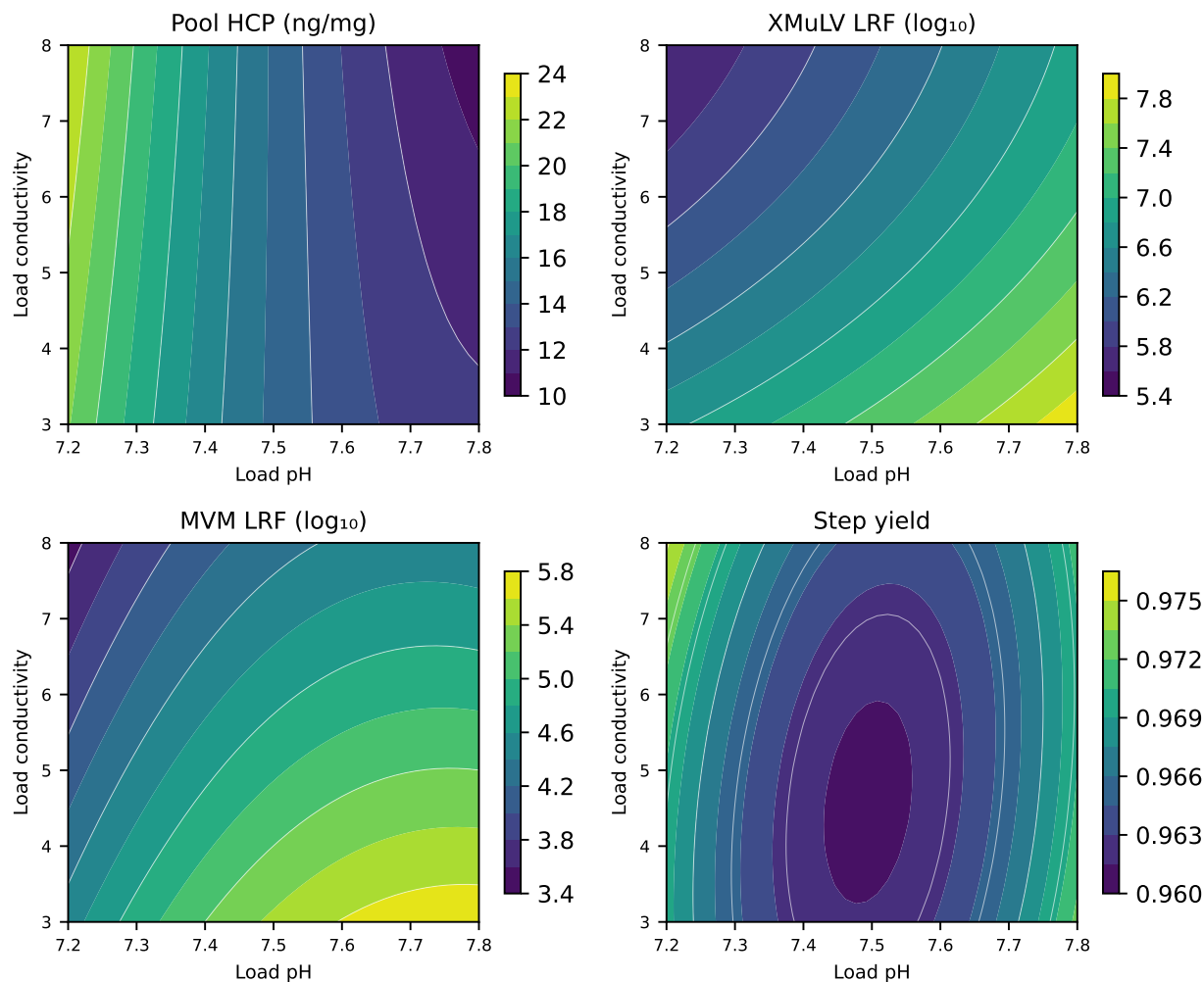


Figure 5.2: Fitted response surfaces over load pH and load conductivity, one panel per response.

Model diagnostics for the two attributes that carry the control strategy are given in Figure 5.3 and Figure 5.4. Each figure plots the standardized residuals against the predicted value, the normal quantile plot of those residuals, and the predicted value against the measured value. The residuals of both models scatter about zero with no trend against the predicted value, the quantile plots are close to linear, and the predicted values track the measured ones across the whole range of the response. No run was excluded from either fit.

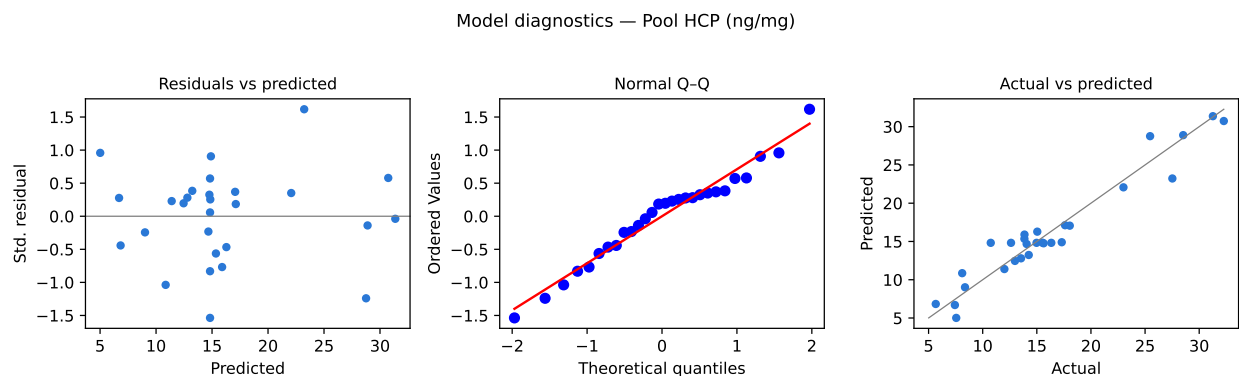


Figure 5.3: Model diagnostics for the pool host cell protein response-surface model.

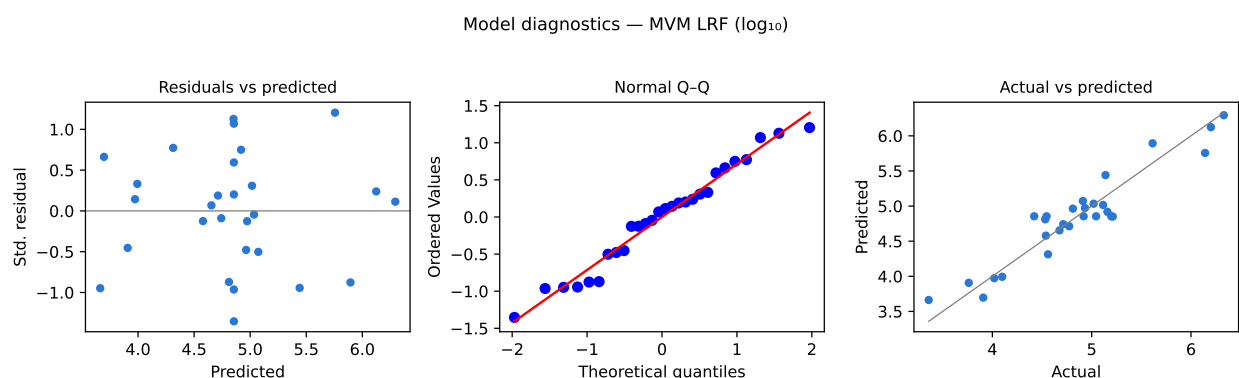


Figure 5.4: Model diagnostics for the MVM log reduction factor response-surface model.

The models predict mean levels of each response over the ranges of the parameters included in them. At an operating point where the predicted mean sits exactly on a limit, about half of the batches run at that point would fall on the wrong side of the limit if the variation about the mean is symmetric. For that reason the proven acceptable ranges in §7 are determined from a predictive interval as well as from the mean, and the design space in §6 is stated on mean predictions and is read together with §7.

5.4 Mechanistic interpretation

Host cell proteins reach the column as a population with a distribution of isoelectric points, and only the fraction that carries a net negative charge at the load pH is held on the quaternary amine ligand. Raising the load pH brings the buffer above the isoelectric point of more of that population, gives those proteins a net negative charge and holds them on the ligand, which is why pool host cell protein falls as load pH rises. Raising the conductivity of the equilibration and wash-1 buffer raises the ionic strength in the bed, screens the attraction between the ligand and the species that

carry the least charge, and releases those species into the pool, which is why pool host cell protein rises with conductivity. A species that carries more negative charge at a high load pH holds its place through a larger rise in ionic strength than a weakly charged one, so the release caused by the buffer conductivity is smaller at high load pH than at low load pH, and the model records that as a negative interaction. No squared term is significant for this response, so no isoelectric point acts as a threshold inside the range studied. A population spread across many isoelectric points gives a surface of that shape, and a single protein with one binding constant would instead give a sharp change at the pH where its charge reverses.

Virus particles carry many acidic groups on their surface and are held on the ligand more tightly than most host cell proteins, and they are captured while the load itself passes through the bed. The ionic strength during the load is the one the load carries, which is why raising load conductivity lowers both log reduction factors. The wash that follows does not lift a bound particle off the ligand, which is why the conductivity of the equilibration and wash-1 buffer appears in neither viral model. Its coefficient is 0.031 log10 per half range for MVM ($p = 0.69$) and -0.076 log10 for XMuLV ($p = 0.32$). A fraction of the host cell protein population is held weakly enough for the wash to displace it, and raising the conductivity of that wash raises the fraction displaced. Prior knowledge expected the wash to release virus particles by the same route, and over the range from 1.6 to 3.6 mS/cm it did not.

The coefficients of load pH and of load conductivity are almost the same for MVM as for XMuLV. Both particles are held by the attraction between the acidic groups on their surface and the quaternary amine ligand, and the lipid envelope that XMuLV carries and MVM lacks takes no part in that attraction. The two differ in the absolute clearance they achieve, and the step is credited with 5.95 log10 for XMuLV and 4.71 log10 for MVM. Removal here is independent of the low-pH hold at Step 6, which disrupts the lipid envelope of XMuLV and leaves the MVM capsid infectious, and independent of the virus filter at Step 9, which retains both particles by size.

Protein load had no significant effect on any response over a range reaching 300 g per L of resin. The ligand binds the impurities and the virus particles and not the antibody, so the capacity that can be exhausted is the capacity for the bound species, and the impurity mass carried onto the column at the top of this range stays far below it. The first execution of this study, run on a load whose acidic charge variant burden was raised, did show a protein load effect on pool host cell protein, and §13.1 gives what caused it.

Step yield showed no dependence on any factor. The antibody carries a net positive charge at every load pH in the range studied, because its isoelectric point lies well above that range, so it is not retained on the ligand and its recovery follows the volumes collected. The one route by which a studied parameter could lower recovery is a load pH high enough for part of the antibody population to approach its isoelectric point and bind, and the upper edge of the range studied, 7.8, stays well below it.

6 Design space

The design space of this step is the part of the characterized region in which every criterion of §3.5 is met. It was evaluated by predicting the 3 responses that carry an acceptance criterion from the response-surface models, on an evenly spaced grid over the four characterized ranges, at 14,641 operating points in total, and testing each point against every criterion.

Of the characterized region, 85.6% meets every criterion. The host cell protein criterion excludes 14.4% of the region and the two viral criteria exclude no part of it, so within the ranges studied the step delivers its required contribution to both viral claims everywhere, including at the corner least favourable to clearance. At the lowest load pH and the highest load conductivity of the characterized region the model predicts 3.53 log₁₀ for MVM against a required 3.28 log₁₀, and 5.55 log₁₀ for XMuLV against a required 3.78 log₁₀, with the other two parameters at their set-points.

The boundary of the design space therefore lies where pool host cell protein meets its in-process limit, and it lies in the plane of load pH against the equilibration and wash-1 conductivity. Figure 5.1 shows that plane. The excluded corner is the one at low load pH and high conductivity, where the host cell proteins carry the least negative charge and the ions in the buffer screen what charge they carry. At the set-point the model predicts 14.8 ng/mg against an in-process limit of 21.7 ng/mg. At the extreme corner of the characterized region, with load pH at 7.2 and the equilibration and wash-1 conductivity at 3.6 mS/cm, it predicts 31.8 ng/mg, which is above that limit.

Three bounds apply to this design space. It is defined on mean predictions from the response-surface models, so a point on its boundary is a point at which half the batches would exceed the limit, and §7 gives the ranges that survive a predictive interval. It is defined over the four parameters of the multivariate design and no other, so it says nothing about a parameter held at its platform value in every run. It is defined on the scale-down model, and design space boundaries are not confirmed at commercial scale at the edges of the ranges.

The normal operating ranges of Table 4.1 sit inside the characterized region on every axis. For load pH, the equilibration and wash-1 conductivity and load conductivity the routine process is operated near the middle of the characterized range, and for protein load the normal operating range reaches closer to the upper edge. Operation anywhere inside the design space is expected to meet every criterion, and under ICH Q8(R2) movement within it is not regarded as a change for filing purposes (International Council for Harmonisation 2009). Movement is still assessed prospectively under the change management system of the pharmaceutical quality system, as ICH Q10 requires (International Council for Harmonisation 2008). Operation outside the design space requires an assessment against this report and against the in-process results of the affected batch.

7 Proven acceptable ranges

A proven acceptable range is the part of a parameter's characterization range over which the attribute in question stays within the criterion that applies to it, with the other parameters of the design accounted for. It is a statement about one parameter and one attribute at a time, and it is reported here for every combination of the 4 multivariate parameters with the 3 attributes the step is judged on.

Two analyses were run for each combination. The first scans the parameter across its characterization range with the other three parameters of the design held constant, and reports the range over which the predicted mean meets the criterion. The second varies the other three parameters within their normal operating ranges by Monte-Carlo sampling of the fitted model, adds the residual variation of the model to each draw, and reports the range over which the 95 % predictive interval of the attribute stays within the criterion. The second analysis is the more demanding of the two, because it requires the attribute to hold not only on average but across the spread the rest of the operating point produces.

Table 7.1 gives both analyses for every parameter and attribute, together with the criterion each was measured against. The criterion for pool host cell protein is the in-process limit of \$3.5. The criteria for the two viral responses are the step contributions the cumulative claims require.

Table 7.1: Proven acceptable ranges, by attribute and parameter.

CQA	Parameter	Char. range	Unit	Crite- rion	PAR (set-point)	PAR (NOR)
Pool HCP (ng/mg)	Load pH	7.2-7.8	pH	≤ 21.7	7.21-7.8	7.45-7.8
Pool HCP (ng/mg)	Equil/Wash-1 conductivity	1.6-3.6	mS/cm	≤ 21.7	1.6-3.45	1.6-2.8
Pool HCP (ng/mg)	Load conductivity	3-8	mS/cm	≤ 21.7	3-8	3-8
Pool HCP (ng/mg)	Protein load	50-300	g/L resin	≤ 21.7	50-300	50-300
XMuLV LRF ($\log_{10}$)	Load pH	7.2-7.8	pH	≥ 3.78	7.2-7.8	7.2-7.8
XMuLV LRF ($\log_{10}$)	Equil/Wash-1 conductivity	1.6-3.6	mS/cm	≥ 3.78	1.6-3.6	1.6-3.6
XMuLV LRF ($\log_{10}$)	Load conductivity	3-8	mS/cm	≥ 3.78	3-8	3-8
XMuLV LRF ($\log_{10}$)	Protein load	50-300	g/L resin	≥ 3.78	50-300	50-300

CQA	Parameter	Char. range	Unit	Crite- rion	PAR (set-point)	PAR (NOR)
MVM LRF (log ₁₀)	Load pH	7.2-7.8	pH	≥ 3.28	7.2-7.8	7.2-7.8
MVM LRF (log ₁₀)	Equil/Wash-1 conductivity	1.6-3.6	mS/cm	≥ 3.28	1.6-3.6	1.6-3.6
MVM LRF (log ₁₀)	Load conductivity	3-8	mS/cm	≥ 3.28	3-8	3-8
MVM LRF (log ₁₀)	Protein load	50-300	g/L resin	≥ 3.28	50-300	50-300

For the two viral responses every parameter is acceptable across its whole characterization range under both analyses. This follows from the size of the margin. The smallest predicted MVM clearance anywhere in the characterized region is well above the contribution the cumulative claim requires of this step, and the predictive interval that the Monte-Carlo analysis adds is narrow beside that margin.

For pool host cell protein two parameters are bounded. Load pH is acceptable over 7.21-7.8 under the first analysis, and the range narrows to 7.45-7.8 under the second. The equilibration and wash-1 conductivity is acceptable over 1.6-3.45 mS/cm under the first analysis and over 1.6-2.8 mS/cm under the second. Load conductivity and protein load are acceptable across their whole ranges under both.

The second analysis therefore excludes part of the normal operating range for these two parameters. The normal operating range of load pH runs from 7.35 to 7.65, and its lower part lies below the proven acceptable range under the second analysis. The normal operating range of the equilibration and wash-1 conductivity runs from 2.1 to 3.1 mS/cm, and its upper part lies above. Both exclusions come from the same corner of the operating region, where the host cell proteins carry the least negative charge and the ions in the buffer screen what charge they carry. With load pH at the bottom of its normal operating range and the equilibration and wash-1 conductivity at the top of its own, and load conductivity and protein load at their set-points, the model predicts 21.9 ng/mg against the in-process limit of 21.7 ng/mg. At the set-point of both parameters the same model predicts 14.8 ng/mg.

Three things bound what that exclusion means. The in-process limit is not the drug substance criterion. It sits a factor below the drug substance criterion of 100 ng/mg, so a pool at the limit still yields drug substance inside that criterion. The nominal process operates at 18.9 ng/mg in the pool, which is well below the limit. The drug substance capability for host cell protein is 6.14 (§8). The corner is nonetheless the part of the operating region with the least margin, and §10 states the control that follows from it.

Figure 7.1, Figure 7.2 and Figure 7.3 show one proven acceptable range for each of the three attributes, plotted against the parameter with the largest main effect on it. Each figure has two panels, the first analysis on the left and the second on the right. The acceptance limit is drawn as a dashed line, the acceptable region of the parameter is shaded green, the normal operating range is shaded grey and the

set-point is marked. The right-hand panel adds the median and the 95 % predictive interval of the Monte-Carlo analysis.

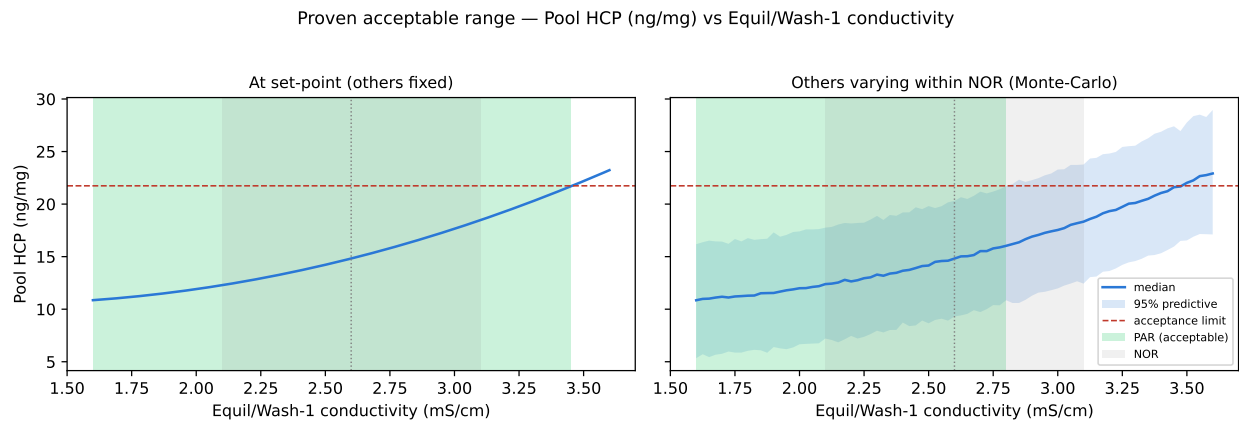


Figure 7.1: Proven acceptable range of the equilibration and wash-1 conductivity for pool host cell protein.

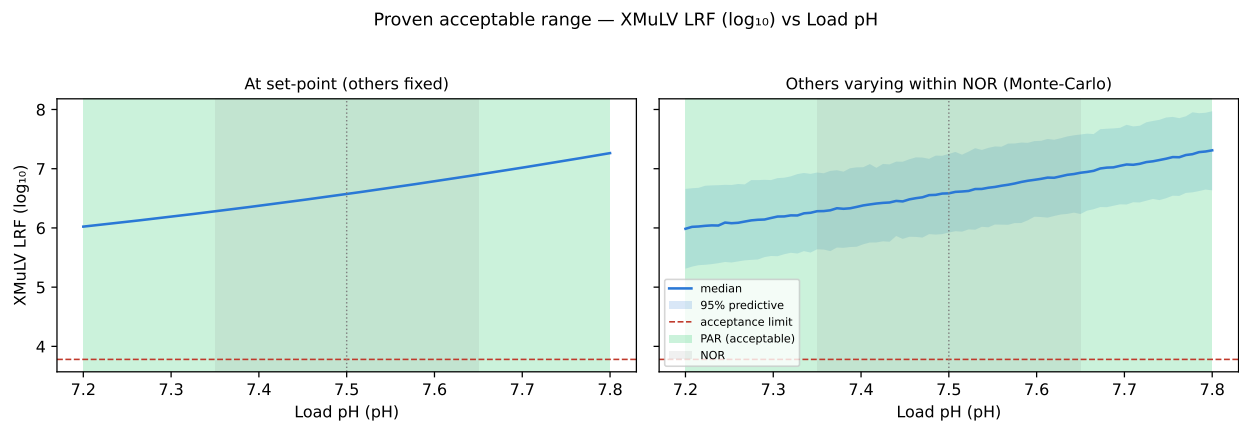


Figure 7.2: Proven acceptable range of load pH for the XMuLV log reduction factor.

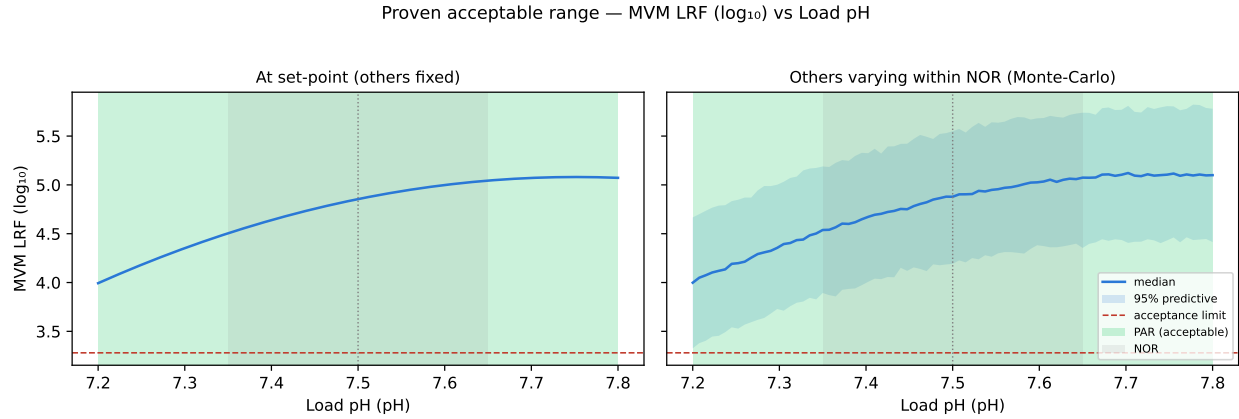


Figure 7.3: Proven acceptable range of load pH for the MVM log reduction factor.

The proven acceptable ranges and the design space answer different questions and are read together. A proven acceptable range is a one-dimensional statement, useful when a single parameter moves and the rest of the operating point does not. The design space of §6 is the multivariate region, and it is what applies when two parameters move at once. The two agree on which attribute bounds this step. Host cell protein bounds both and the viral responses bound neither, and in both analyses the parameters it bounds are load pH and the equilibration and wash-1 conductivity.

8 Process capability and robustness

Capability was estimated at commercial scale by the Monte-Carlo simulation described in §3.5, over 2,000 simulated batches of the whole train. Table 8.1 gives the attributes this step sets or clears.

Table 8.1: Commercial-scale capability of the attributes this step sets or clears.

CQA	Crit.	Spec	Acceptance	Mean $\pm$ SD	Cpk
Host Cell Protein (HCP)	M-H	upper	0–100	15.2 $\pm$ 4.6	6.1
Residual DNA	VL	upper	0–0.001	0.0001 $\pm$ 0	10.8
Leached Protein A	L-M	upper	0–5	0.003 $\pm$ 0.0006	2,785.3
Viral clearance — XMuLV (enveloped)	VH	lower	16.7–30	19.5 $\pm$ 0.61	1.5
Viral clearance — MVM (parvovirus)	VH	lower	8.6–20	10.4 $\pm$ 0.4	1.5

The cumulative MVM clearance has the tightest capability of any A-Mab drug substance attribute, at Cpk 1.51. The simulated batches average 10.38 log₁₀ against a requirement of 8.6 log₁₀, with a standard deviation of 0.40 log₁₀, so the margin is 4.5 standard deviations. The cumulative XMuLV clearance is next, at Cpk 1.53. Both requirements come from the viral safety assessment and not from what the process can deliver. Each cumulative figure also sums three steps, so its spread carries the variability of all three.

Host cell protein is capable at Cpk 6.14, with a mean of 15.2 ng/mg against a criterion of 100 ng/mg. Residual DNA is capable at Cpk 10.81. The leached Protein A statistic in Table 8.1 carries no information. The simulated mean is 0.003 ppm against a criterion of 5 ppm, and at that distance the index measures the spread of the simulation and says nothing about how close the process comes to the limit.

This step contributes to each of these attributes without determining any of them alone. In the nominal process it clears host cell protein 6.2-fold, from 117.0 ng/mg in the cation exchange eluate to 18.9 ng/mg in its own pool, and the Protein A and cation exchange steps carry the clearance before it. It contributes 4.71 log₁₀ of the cumulative MVM claim of 10.03 log₁₀, and 5.95 log₁₀ of the cumulative XMuLV claim of 18.87 log₁₀. The remainder of the MVM claim comes from small-virus retentive filtration (PCR-009) and the remainder of the XMuLV claim from low-pH inactivation (PCR-006) and the same filter. The consolidated ledger is in PCMR-001.

Three bounds apply to the capability figures. They are estimated from qualified scale-down models with every parameter drawn inside its normal operating range, so they describe a process operated as intended and not a process in excursion. They are estimates from a model and are confirmed at commercial scale during Stage 2, which has not been executed. And the viral figures are cumulative over three steps, so a shortfall at any one of them moves the attribute this step is registered against.

The step is robust in the sense that matters for a flow-through polishing step. Yield showed no significant dependence on any studied parameter in either design, and the nominal step yield is 97.3%. Both viral clearances meet their required contributions over the entire characterized region. The one attribute with a boundary inside the characterized region is pool host cell protein, and that boundary is the in-process limit, which sits a factor below the drug substance criterion of 100 ng/mg.

9 Parameter classification

Every parameter was classified under SOP-4001 from the effects demonstrated in this study and from the risk of the parameter leaving its range in routine operation. A parameter whose variation affects a critical quality attribute is a critical process parameter or a well-controlled critical process parameter, and what separates the two is the risk of the parameter falling outside the design space and not the size of its effect. A parameter with no demonstrated link to a quality attribute, which is monitored because it keeps the process consistent, is a key process parameter. Table 4.1 carries the classification of each parameter and the reasons are given here.

Load pH changes the charge carried by the acidic host cell proteins and by the virus particles, and it has the largest main effect on pool host cell protein and on both log reduction factors. It was classified as a well-controlled critical process parameter. The load is titrated against a measured pH before it is applied and holds within a narrow band of its target in routine operation, so the risk of operating outside the characterized range is low.

The equilibration and wash-1 conductivity has the largest effect on pool host cell protein of any parameter studied and is the parameter that bounds the operating region. It was classified as a well-controlled critical process parameter, because it is fixed by buffer preparation and verified against the buffer release specification before the buffer reaches the column, so the risk of operating outside the characterized range is low. The lot described in §13.3 was found by that testing before use.

Load conductivity carries the ions that screen the attraction between the virus particles and the ligand during the load, and both log reduction factors fall as it rises. Pool host cell protein does not move with it. It was classified as a well-controlled critical process parameter, because the conductivity of the load is fixed by the cation exchange eluate and its adjustment, both of which are controlled by the preceding step and measured before the load is applied.

Protein load showed no significant effect on any response over a range reaching 300 g per L of resin. It was nonetheless classified as a well-controlled critical process parameter. The ligand holds a finite mass of bound species, and a load that carries enough impurity to approach that mass lets host cell protein pass into the pool. The first, invalidated execution of this study, run on material with a raised acidic charge variant burden, reached 149.5 ng/mg at the top of the load range against an in-process limit of 21.7 ng/mg (§13.1). The representative material studied here carries too little bound impurity for that effect to appear within the range, and protein load is therefore controlled as a parameter whose variation can reach a critical quality attribute.

Operating flow rate was assessed one factor at a time and was not a factor of either design, so no effect estimate is reported for it. It was classified as a well-controlled

critical process parameter on the basis of the univariate assessment, and it sits with the other four because the pump holds the flow rate close to its set-point against a measured value, which keeps the residence time inside the range assessed.

All 5 parameters of this step are therefore well-controlled critical process parameters and none is a critical process parameter. The whole train carries 1 critical process parameter, the inactivation pH of the low-pH hold (PCR-006), where the range is narrow and the consequence of falling below it is immediate. Nothing at this step has that character. The classification counts for the campaign are consolidated in PCMR-001.

10 Contribution to the control strategy

The step contributes five elements to the control strategy of the drug substance process.

The first is the set of parameter ranges in Table 4.1. The four multivariate parameters are operated within the normal operating ranges given there, and those ranges lie inside the region characterized by this study. The ranges are carried into the batch record and into the Stage 2 protocol.

The second is in-process control of the pool. Pool host cell protein is measured on every batch under AMV-3012 and judged against the in-process limit of \$3.5. That measurement is the control that covers the corner identified in §7, where load pH at the bottom of its normal operating range and the equilibration and wash-1 conductivity at the top of its own leave the least margin to the limit. Pool protein concentration and step yield are recorded on every batch as process consistency measures.

The third is the pool collection method. The trailing-edge stop set-point was corrected from 180 mAU to 120 mAU following the investigation in §13.2, and the corrected value is the one specified for commercial operation.

The fourth is a control on the feed to the step, which follows from the investigation in §13.1. The hold time of the neutralized cation exchange eluate before it is loaded onto this step is specified and recorded, and the 36 h hold that produced the non-representative material lies outside what the specification allows. The charge-variant profile of the load is verified by icIEF under AMV-3013 when a hold is extended beyond that specification.

The fifth is the set of life-cycle controls. Column packing is verified by plate count and peak asymmetry, and resin cycling, sanitization and replacement follow SOP-2008. The models in this report were fitted on resin within the early part of its life cycle, so cumulative cycle number is controlled by that programme and not by the ranges reported here.

Attributes that this step does not control are controlled elsewhere and are named here so that the strategy can be read as a whole. Aggregate is controlled at cation exchange (PCR-007). The glycan and charge-variant attributes are formed and controlled in the production bioreactor (PCR-003). Enveloped virus is inactivated at Step 6 (PCR-006) and both model viruses are retained at Step 9 (PCR-009). Residual DNA and leached Protein A are cleared at this step and at the two chromatographic steps before it, and are tested on the drug substance. The combination of these contributions is consolidated in PCMR-001.

11 Discussion

Two removals share one ligand at this step and answer to different parameters. Host cell protein removal falls when the conductivity of the equilibration and wash-1 buffer rises, because part of the population is held weakly enough for the wash to displace it. Virus removal does not fall, because a virus particle presents many acidic groups to the ligand at once and stays bound through the wash. Prior knowledge treated the ionic strength in the bed as acting on everything held on the ligand, and this study confines that expectation to the weakly held fraction. The operating region is bounded in the plane of load pH against the equilibration and wash-1 conductivity for the same reason, and it runs to the edge of the characterized range in load conductivity and in protein load.

The scale-down model supports these conclusions for the reasons given in §3.1, and the centre-point reproducibility in §5.1 puts a number on how well it repeats. The two designs gave centre-point means that agree within their dispersion. Pool host cell protein and the two log reduction factors gave models with no significant lack of fit against that reproducibility.

Three deviations were recorded and all three are closed. One of them invalidated the first execution of both designs and forced a full re-execution on requalified material. The re-executed study is the study reported here, and the first execution is retained only as evidence for the root cause. The other two were a permissive pool collection set-point, which was corrected, and a buffer released below its pH acceptance range on one screening run, which was retained. §13 gives each in full.

Four limitations bound what this report establishes. The models are fitted on a scale-down column and design space boundaries are not confirmed at commercial scale at the edges of the ranges. The viral clearance figures come from a small-scale spiking model operated under containment, which is the accepted basis for a modular claim under ICH Q5A(R2) (International Council for Harmonisation 2023a) but is not the commercial column. The models are mean-level models, so a point on the boundary of the design space is a point at which about half of batches would exceed the limit, and this is why the proven acceptable ranges of §7 are also computed from a predictive interval. And the whole study was run on resin within the early part of its life cycle, so the effect of resin age on clearance is managed by SOP-2008 and not by these models.

The tightest capability in the drug substance is the cumulative MVM clearance, at Cpk 1.51, and this step is one of the two that carry it. A future change to load pH or to load conductivity, even inside the characterized region, is therefore assessed against the cumulative claim in PCMR-001 and not against this step alone.

12 Conclusions

The anion exchange step was characterized on a qualified scale-down model over ranges wider than its normal operating ranges. Of its 5 parameters, 4 were studied in a 19-run screening design and a 28-run response-surface design, and 1 was assessed one at a time.

Pool host cell protein falls as load pH rises and rises as the conductivity of the equilibration and wash-1 buffer rises, because the first holds the acidic proteins on the ligand and the second screens the charge that holds them. Both log reduction factors fall as load pH falls and as load conductivity rises, because virus particles are bound by the acidic groups on their surface while the load passes. No studied parameter changes step yield, and the antibody is not retained on the ligand at any load pH in the range. The response-surface models of these three attributes are adequate, show no significant lack of fit, and are the predictive models of this report.

Of the characterized region, 85.6% meets every criterion the step is judged against, and pool host cell protein is the attribute that bounds it. Proven acceptable ranges are reported for every parameter against every attribute that carries an acceptance criterion at this step. The two viral clearances are acceptable across every characterization range under both analyses, and pool host cell protein bounds load pH and the equilibration and wash-1 conductivity when the rest of the operating point is allowed to vary within its normal operating ranges.

At commercial scale the attributes this step sets or clears are capable, and the tightest is the cumulative MVM clearance at Cpk 1.51, which this step shares with small-virus retentive filtration. All 5 parameters were classified as well-controlled critical process parameters.

The study recorded 3 deviations, and all of them were investigated and closed. The first execution of both designs was invalidated by a non-representative load and re-executed in full, and the re-executed study is what this report analyses. The ranges, the in-process controls and the classifications reported here are proposed for Stage 2 and roll up into PCMR-001.

13 Deviations from the plan

Execution of the study recorded 3 deviations from PCP-008. Table 13.1 lists them with the activity that detected each one and its disposition, and the subsections that follow give the investigation and the impact assessment of each.

Table 13.1: Deviation register for this study.

De- via- tion	Summary	Detected during	Disposition
DEV-008-01	Non-representative (deamidated) load in the first DoE execution; designs invalidated and re-executed	load-material charge-variant verification by icIEF (AMV-3013)	invalidated_and_re_executed
DEV-008-02	Trailing-edge UV pool-collection set-point slightly permissive in both executions	post-execution pool-fraction reconciliation (AMV-3012)	corrected_by_modelling_and_verification_runs
DEV-008-03	Equilibration/Wash-1 buffer released below its pH acceptance range on one screening run	buffer release testing per SOP-2103	retained

13.1 DEV-008-01 — non-representative load in the first execution

The first execution of both designs was run on a load whose charge-variant profile was not representative of the routine process. Verification of the load material by icIEF under AMV-3013 measured acidic charge variants at 33.5 % of the total, against 21 % in the requalified material used for the reported study. The affected lot was LOT-CEX-8842.

The investigation traced the elevated acidic fraction to the hold of the neutralized cation exchange eluate, which ran 36 h before the material was loaded. An extended hold at neutral pH promotes asparagine deamidation, and deamidation converts asparagine to aspartate, which lowers the isoelectric point of the affected molecules and moves them into the acidic fraction. The root cause was therefore the hold time and not the step, and the material entering the column was not the material the commercial process delivers.

The consequence for the first execution was a load-dependent loss of host cell protein clearance. In the screening data of that execution, protein load interacted with the equilibration and wash-1 conductivity with an effect of 34.8 ng/mg ($p = 0.002$), and protein load itself carried a main effect of 35.6 ng/mg ($p = 0.002$). Pool host cell protein reached 149.5 ng/mg at the corner of the design combining the highest protein load with the highest conductivity and the lowest load pH, well above the in-process limit of 21.7 ng/mg. A deamidated antibody carries more negative charge than the routine product, so a fraction of it bound the ligand instead of flowing through. At a high protein load that fraction occupied ligand the host cell proteins would otherwise have occupied, and the higher wash conductivity then released what little host cell protein remained bound.

Both designs were declared invalid for the definition of the operating region and were re-executed in full on requalified, representative material, the cation exchange eluate LOT-CEX-8851, whose charge-variant profile was verified before execution. The requalified data confirm the root cause. The same interaction between protein load and the equilibration and wash-1 conductivity is 0.22 ng/mg ($p = 0.86$) in the reported screening design, which is smaller than the standard error of the estimate, and the highest pool host cell protein measured anywhere in that design is 35.7 ng/mg. The models fitted to the two executions differ accordingly. At the highest protein load and the highest equilibration and wash-1 conductivity of the design, the first execution predicts 78.8 ng/mg and the reported execution predicts 21.1 ng/mg.

The first execution is retained as a superseded dataset and is used in this report only to confirm the root cause. No effect estimate, model, proven acceptable range or classification in this report is derived from it. The deviation is closed as invalidated and re-executed, and the control that prevents its recurrence is the hold-time specification described in §10.

13.2 DEV-008-02 — permissive pool collection set-point

The trailing-edge stop set-point for pool collection was 180 mAU in both executions of the study, which is more permissive than the method intends. Reconciliation of the pool fractions after execution, using the host cell protein results of AMV-3012, showed that pool collection continued further into the trailing edge than specified, so each pool carried a small additional volume of late material.

The set-point applied identically to every run of both designs, so it shifted every pool in the same direction by about the same amount. An estimated effect is a difference between runs, and a shift shared by all of them cancels out of every difference, so the effects, coefficients and interactions reported in §5 are unaffected. What the shift can bias is the absolute level of pool host cell protein, and the extra late material it collected raises that level, so the operating region in §6 and the ranges in §7 are narrower than the corrected method would give.

The set-point was corrected to 120 mAU. The pool composition at the corrected set-point was modelled from the collected fractions, and the correction was confirmed by 3 verification runs at the corrected set-point, 2 of them at the edges of the operating

range. That verification set has a limit. With the precision of AMV-3012 and 3 runs, a difference smaller than 23.6 % of the mean cannot be resolved at 95 % confidence, so the verification bounds the effect of the correction without measuring it exactly. The deviation is closed, and the corrected set-point is carried into the control strategy (§10).

13.3 DEV-008-03 — equilibration and wash-1 buffer released below its pH range

The equilibration and wash-1 buffer lot LOT-BUF-8815 was released at pH 7.32 against a lower acceptance limit of 7.4 and a target of 7.5. The lot was used on one factorial run of the screening design. The excursion was found by buffer release testing under SOP-2103, and the root cause was a titration endpoint error at buffer preparation.

The impact is bounded on three grounds. The affected run is one factorial run of the screening design, and this report uses the screening design to identify which factors act and not to predict a response. The response-surface design, which carries every prediction, every proven acceptable range and the operating region, was executed entirely on buffer released within its acceptance range. And a buffer pH below target leaves the acidic host cell proteins less negatively charged in the bed, so it would raise pool host cell protein in that run and could not have made the step look better than it is.

The run was retained in the screening analysis and the deviation is closed as retained. Buffer release testing under SOP-2103 catches a preparation error of this kind before the buffer reaches the column, and it did so here.

Appendix A — Screening design matrix and responses

Table 13.2 gives the executed screening design in coded units with the measured responses. The factor letters are those of Table 4.2, the coded levels -1 and $+1$ are the edges of each characterization range and 0 is the midpoint of that range, and the run column is the design position and not the execution order, which was randomized.

Table 13.2: Screening design in coded units, with measured responses.

Run	Type	A	B	C	D	Pool HCP (ng/mg)	XMuLV LRF (log ₁₀)	MVM LRF (log ₁₀)	Step yield
1	factorial	-1	-1	-1	-1	12.5	6.16	4.59	0.99
2	factorial	-1	-1	-1	1	15.6	6.15	4.43	0.975
3	factorial	-1	-1	1	-1	12.1	5.47	3.76	0.951
4	factorial	-1	-1	1	1	14.6	5.56	3.96	0.964
5	factorial	-1	1	-1	-1	35.7	6.34	4.74	0.965
6	factorial	-1	1	-1	1	35	6.3	4.83	0.995
7	factorial	-1	1	1	-1	29.2	5.92	4.01	0.974
8	factorial	-1	1	1	1	29.8	5.95	3.94	0.969
9	factorial	1	-1	-1	-1	9.62	7	5.91	0.973
10	factorial	1	-1	-1	1	7.37	7.4	5.61	0.953
11	factorial	1	-1	1	-1	8.69	6.61	5.25	0.972
12	factorial	1	-1	1	1	8.04	6.71	4.89	0.973
13	factorial	1	1	-1	-1	12.7	6.59	5.91	0.995
14	factorial	1	1	-1	1	15	8.34	5.97	0.981

Run	Type	A	B	C	D	Pool HCP (ng/mg)	XMuLV LRF (log ₁₀)	MVM LRF (log ₁₀)	Step yield
15	facto- rial	1	1	1	-1	12.9	6.69	5.15	0.966
16	facto- rial	1	1	1	1	15	6.52	5.31	0.965
17	center	0	0	0	0	16.2	6.42	4.54	0.966
18	center	0	0	0	0	17.5	6.8	5.08	0.975
19	center	0	0	0	0	11.6	5.95	4.83	0.974

Appendix B — Response-surface design matrix and responses

Table 13.3 gives the executed response-surface design in the same form. The axial runs sit at the face of the coded cube, so every level in the table is at an edge or at the midpoint of a characterization range.

Table 13.3: Response-surface design in coded units, with measured responses.

Run	Type	A	B	C	D	Pool HCP (ng/mg)	XMuLV LRF (log ₁₀)	MVM LRF (log ₁₀)	Step yield
1	factorial	-1	-1	-1	-1	13.5	6.72	4.55	0.951
2	factorial	-1	-1	-1	1	17.3	6.03	5.21	0.989
3	factorial	-1	-1	1	-1	14.1	5.87	4.02	0.995
4	factorial	-1	-1	1	1	15	5.3	3.76	0.951
5	factorial	-1	1	-1	-1	28.5	6.53	4.77	0.988
6	factorial	-1	1	-1	1	25.5	6.71	4.71	0.963
7	factorial	-1	1	1	-1	31.3	5.36	3.91	0.995
8	factorial	-1	1	1	1	32.3	5.06	3.36	0.966
9	factorial	1	-1	-1	-1	5.66	7.5	6.14	0.974
10	factorial	1	-1	-1	1	8.37	7.83	5.61	0.989
11	factorial	1	-1	1	-1	7.56	6.49	4.54	0.972
12	factorial	1	-1	1	1	7.43	7.13	4.68	0.988
13	factorial	1	1	-1	-1	17.6	7.18	6.2	0.974
14	factorial	1	1	-1	1	18	8.12	6.33	0.968
15	factorial	1	1	1	-1	13.9	6.58	4.53	0.96

Run	Type	A	B	C	D	Pool HCP (ng/mg)	XMuLV LRF (log ₁₀)	MVM LRF (log ₁₀)	Step yield
16	facto- rial	1	1	1	1	13.9	6.37	5.16	0.973
17	axial	-1	0	0	0	23	5.83	4.1	0.971
18	axial	1	0	0	0	12	7.37	4.91	0.965
19	axial	0	-1	0	0	8.1	6.42	4.93	0.946
20	axial	0	1	0	0	27.5	6	5.02	0.972
21	axial	0	0	-1	0	15.6	7.44	5.14	0.952
22	axial	0	0	1	0	15.5	5.93	4.56	0.967
23	axial	0	0	0	-1	13	6.52	4.81	0.968
24	axial	0	0	0	1	14.3	6.7	5.11	0.952
25	center	0	0	0	0	15	6.59	4.42	0.958
26	center	0	0	0	0	10.7	6.03	5.2	0.975
27	center	0	0	0	0	16.3	7.05	5.04	0.977
28	center	0	0	0	0	12.6	6.86	4.92	0.953

Appendix C — Full effect and coefficient tables

The screening effect tables for pool host cell protein and for MVM are in §5.2, and the response-surface coefficient tables for pool host cell protein, MVM and XMuLV are in §5.3. This appendix gives the remaining tables. Table 13.4 and Table 13.5 give the screening effects for XMuLV and for step yield, and Table 13.6 gives the response-surface coefficients for step yield.

Table 13.4: Screening effect estimates, XMuLV log reduction factor.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	1	0.501	0.0982	5.1	0.000934	***
C	-0.605	-0.302	0.0982	-3.08	0.0151	*
D	0.268	0.134	0.0982	1.37	0.209	
CD	-0.255	-0.128	0.0982	-1.3	0.23	
AD	0.25	0.125	0.0982	1.27	0.24	
B	0.198	0.0991	0.0982	1.01	0.342	
BD	0.126	0.0632	0.0982	0.643	0.538	
AC	-0.0968	-0.0484	0.0982	-0.493	0.635	
AB	-0.094	-0.047	0.0982	-0.478	0.645	
BC	-0.0169	-0.00844	0.0982	-0.0859	0.934	

Table 13.5: Screening effect estimates, step yield.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
C	-0.0116	-0.00582	0.00315	-1.85	0.102	
B	0.00743	0.00371	0.00315	1.18	0.272	
AD	-0.00715	-0.00357	0.00315	-1.13	0.29	
AC	0.00513	0.00257	0.00315	0.815	0.439	
BC	-0.00392	-0.00196	0.00315	-0.622	0.551	
BD	0.00383	0.00192	0.00315	0.608	0.56	
CD	0.00346	0.00173	0.00315	0.549	0.598	
AB	0.00141	0.000707	0.00315	0.224	0.828	
D	-0.00136	-0.000679	0.00315	-0.215	0.835	
A	-0.000663	-0.000332	0.00315	-0.105	0.919	

Table 13.6: Response-surface coefficients, step yield.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.961	0.00501	192	7.92e-24	***
A	-0.000294	0.00342	-0.086	0.933	
B	0.00022	0.00342	0.0643	0.95	
C	0.00109	0.00342	0.318	0.756	
D	-0.00203	0.00342	-0.594	0.563	
AB	-0.00464	0.00362	-1.28	0.223	
AC	-0.00175	0.00362	-0.482	0.637	
AD	0.00608	0.00362	1.68	0.117	
BC	-0.000152	0.00362	-0.042	0.967	
BD	-0.00433	0.00362	-1.2	0.253	
CD	-0.00413	0.00362	-1.14	0.275	
A ²	0.00932	0.00902	1.03	0.32	
B ²	0.000524	0.00902	0.0581	0.955	
C ²	0.00152	0.00902	0.168	0.869	
D ²	0.00177	0.00902	0.196	0.847	

Appendix D — Analytical methods summary

Table 13.7 lists the validated methods used to measure the responses and the load material of this study, with the analyte each measures and the form in which each result is reported. Table 13.8 gives the validated performance of those methods for which a performance record is held, as intermediate precision, limit of quantitation, accuracy and the share of the total observed variance attributable to the method.

Table 13.7: Validated analytical methods used in this study.

Reference	Technique	Analyte	Reported as
AMV-3012	Enzyme-linked immunosorbent assay	Host cell protein	ng per mg of product
AMV-3013	Imaged capillary isoelectric focusing	Charge variants of the load	% of total peak area
AMV-3014	Quantitative polymerase chain reaction	Residual DNA	ng per dose
AMV-3016	Enzyme-linked immunosorbent assay	Leached Protein A	ppm
AMV-3017	Infectivity titre by TCID50	XMuLV	log10 TCID50 per mL
AMV-3018	Infectivity titre by TCID50 with qPCR confirmation	MVM	log10 TCID50 per mL

Table 13.8: Validated performance of the analytical methods.

Refer- ence	Method	Precision (%)	LOQ	LOQ unit	Accuracy (%)	Variance share
AMV-3016	Leached Protein A by ELISA	6.5	0.25	ppm	97.4	0.41
AMV-3017	XMuLV Infectivity Titre (TCID50)	0.3	1.5	log10 TCID50/mL	96	0.28
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	0.32	1.4	log10 TCID50/mL	96	0.3

Refer- ence	Method	Precision (%)	LOQ	LOQ unit	Accuracy (%)	Variance share
AMV- 3012	Host-Cell Protein by ELISA	9.5	1	ng/mg	95	0.4

References

- CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.
- International Council for Harmonisation. 2008. *ICH Q10: Pharmaceutical Quality System*. ICH.
- International Council for Harmonisation. 2009. *ICH Q8(R2): Pharmaceutical Development*. ICH.
- International Council for Harmonisation. 2012. *ICH Q11: Development and Manufacture of Drug Substances*. ICH.
- International Council for Harmonisation. 2023a. *ICH Q5A(R2): Viral Safety Evaluation of Biotechnology Products Derived from Cell Lines of Human or Animal Origin*. ICH.
- International Council for Harmonisation. 2023b. *ICH Q9(R1): Quality Risk Management*. ICH.
- U.S. Food and Drug Administration. 2011. *Guidance for Industry: Process Validation — General Principles and Practices*. FDA.